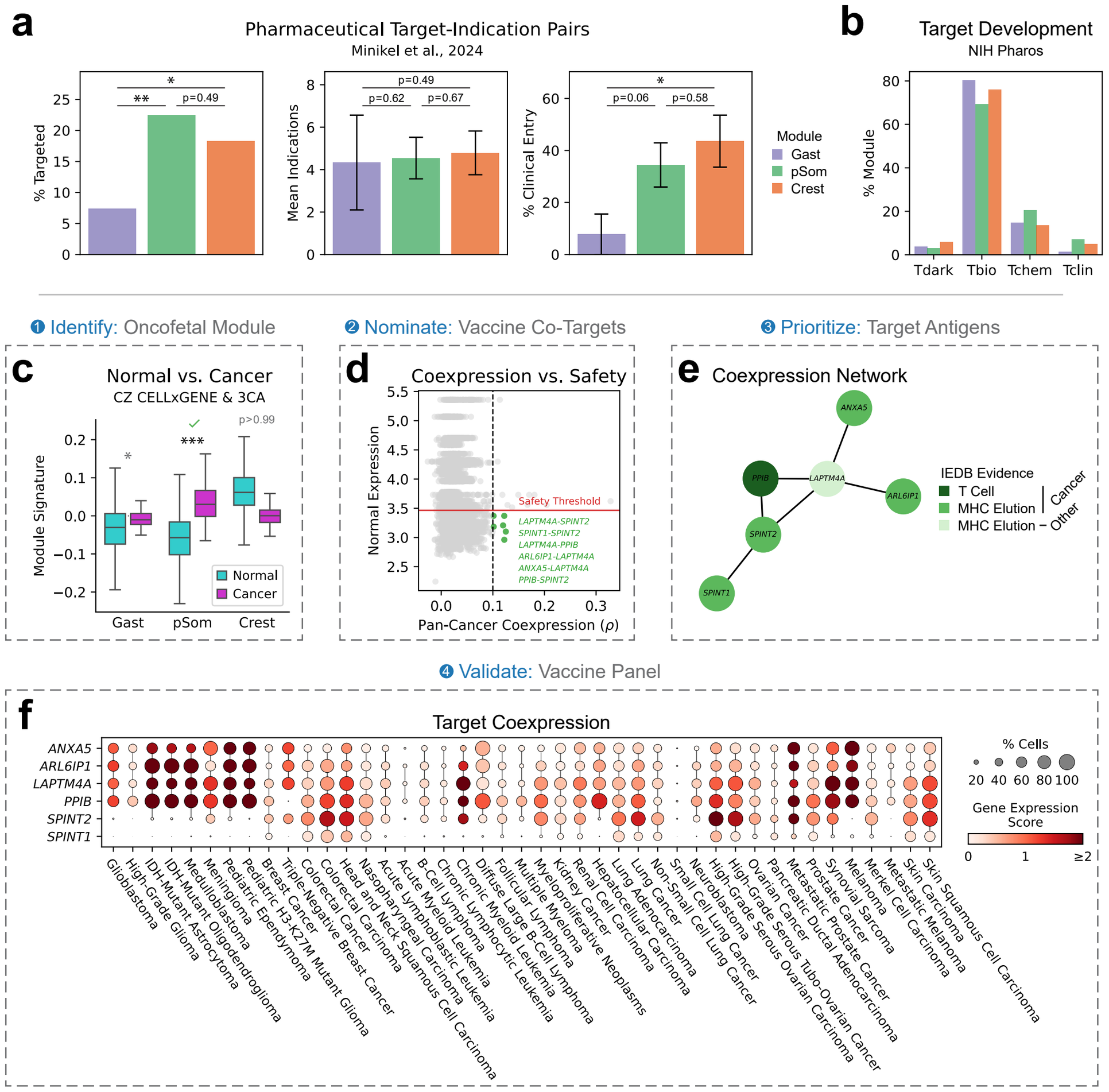

a
Pharmaceutical Target-Indication Pairs
Minikel et al., 2024
% Targeted
Mean Indications
% Clinical Entry
p=0.49
p=0.62
p=0.67
p=0.06
p=0.58
Module
Gast
pSom
Crest
b
Target Development
NIH Pharos
% Module
Tdark
Tbio
Tchem
Tclin
1 Identify: Oncofetal Module
2 Nominate: Vaccine Co-Targets
3 Prioritize: Target Antigens
c
Normal vs. Cancer
CZ CELLxGENE & 3CA
Module Signature
p>0.99
Normal
Cancer
Gast
pSom
Crest
d
Coexpression vs. Safety
Normal Expression
Safety Threshold
LAPTM4A-SPINT2
SPINT1-SPINT2
LAPTM4A-PPIB
ARL6IP1-LAPTM4A
ANXA5-LAPTM4A
PPIB-SPINT2
Pan-Cancer Coexpression (ρ)
e
Coexpression Network
ANXA5
PPIB
LAPTM4A
ARL6IP1
SPINT2
SPINT1
IEDB Evidence
T Cell
MHC Elution
MHC Elution – Other
Cancer
Other
4 Validate: Vaccine Panel
f
Target Coexpression
ANXA5
ARL6IP1
LAPTM4A
PPIB
SPINT2
SPINT1
% Cells
Gene Expression Score
Glioblastoma
High-Grade Glioma
IDH-Mutant Astrocytoma
IDH-Mutant Oligodendroglioma
Medulloblastoma
Meningioma
Pediatric Ependymoma
Pediatric H3-K27M Mutant Glioma
Breast Cancer
Triple-Negative Breast Cancer
Colorectal Cancer
Colorectal Carcinoma
Head and Neck Squamous Cell Carcinoma
Nasopharyngeal Carcinoma
Acute Lymphoblastic Leukemia
Acute Myeloid Leukemia
B-Cell Lymphoma
Chronic Lymphocytic Leukemia
Chronic Myeloid Leukemia
Diffuse Large B-Cell Lymphoma
Follicular Lymphoma
Multiple Myeloma
Myeloproliferative Neoplasms
Kidney Cancer
Renal Cell Carcinoma
Hepatocellular Carcinoma
Lung Adenocarcinoma
Lung Cancer
Non-Small Cell Lung Carcinoma
Small Cell Lung Cancer
Neuroblastoma
High-Grade Serous Ovarian Cancer
High-Grade Serous Tubo-Ovarian Cancer
Ovarian Cancer
Pancreatic Ductal Adenocarcinoma
Metastatic Prostate Cancer
Prostate Cancer
Synovial Sarcoma
Melanoma
Merkel Cell Carcinoma
Metastatic Melanoma
Skin Carcinoma
Skin Squamous Cell Carcinoma